

PEDRO PABLO ZAVALA TEJOS

Computer Science Engineer | Data Engineering and Cloud

pedropablozavalat@uc.cl | (+56) 9 5223 1388 | Santiago, Chile | Portfolio | LinkedIn | GitHub

EDUCATION

Pontificia Universidad Católica de Chile

Civil Engineering in Computer Science

2021 – Jul. 2026
Santiago, Chile

- Major in Computing and Information Systems.
- Minor in Data Science and Analytics.
- Academic average: 6.22/7.00.
- Class rank: 47/950 – Top 5%.

PROFESSIONAL EXPERIENCE

Research Assistant

UC School of Engineering

Mar. 2025 – Present
Santiago / Estación Patagonia UC

- SAVIA and ECHO:** designed and developed a hybrid *edge-to-cloud* infrastructure for remote scientific stations within UC's Regional Field Stations Network, focused on automating the acquisition, validation, backup, and synchronization of telemetry and environmental sensor data. Developed ECHO, a Home Assistant integration, and EchoLib, a Python library for connecting and processing heterogeneous sources through FTP/SFTP, REST APIs, sensors, and cloud storage services.
- PISMA-WP (Avanza UC):** design and develop the backend architecture and ETL workflows of a socio-environmental monitoring platform, integrating satellite, geospatial, and environmental data through reproducible APIs. I also implement its AWS infrastructure using Terraform and Docker.
- SAVIA – ML on the Edge:** develop data preparation, validation, and processing pipelines for camera-trap datasets used in animal detection and species-classification workflows.

Undergraduate Researcher – SCORPIO

IoT-UC Laboratory, UC School of Engineering – Avanza UC

Mar. 2026 – Jun. 2026
Santiago, Chile

- Designed and implemented a modular *edge-to-cloud* architecture to acquire, process, back up, and centralize LoRa satellite data under intermittent connectivity.
- Developed edge services in Python and Docker on a Raspberry Pi using MQTT and SQLite, together with a REST API built with Node.js, TypeScript, and Prisma to manage stations, satellites, and received packets.

Capstone Project – Natural-Language Data Query Assistant

National Center for Artificial Intelligence (CENIA), UC School of Engineering

Mar. 2026 – Jun. 2026
Santiago, Chile

- Developed, as part of a six-member team, a conversational assistant that translates natural-language requests into validated and executable SQL queries, using a cloud architecture that integrates Slack, RAG, PostgreSQL/pgvector, LLM-based validation, and automated deployment on AWS with Terraform.

Data Engineering Intern

SimplePark

Jan. 2025 – Apr. 2025
Remote

- Developed and automated ETL pipelines and event-driven processes for financial reconciliation and operational analytics using Python, SQL, BigQuery, Cloud Functions, Pub/Sub, and Google Cloud Storage, integrating Mercado Pago and Kushki APIs and BSale reporting workflows.
- Refactored data processes following Clean Architecture principles, incorporating validations and unit testing, and designed Looker Studio dashboards that reduced manual tasks from several hours to a few minutes.

Capstone Project – AI-Assisted Academic Management Platform

2024

Faculty of Health Sciences, Pontificia Universidad Católica de Chile

Aug. 2024 – Dec. 2024
Santiago, Chile

- Implemented backend services for an AI-assisted web platform using NestJS, TypeScript, Prisma, and PostgreSQL.
- Integrated external data sources and redesigned relational database schemas to improve system maintainability and scalability.

Teaching Assistant – Advanced Programming (IIC2233)

Pontificia Universidad Católica de Chile

Mar. 2023 – Jul. 2024
Santiago, Chile

- Supported students during programming sessions and evaluated Python assignments and coursework.

PROFESSIONAL SKILLS

Data Engineering: Python, SQL, ETL/ELT, Star Schema, data quality and validation

Processing and Storage: BigQuery, PostgreSQL, MongoDB, Pandas, Google Cloud Storage, SQLite

Cloud and Processing: Google Cloud Platform, AWS, Cloud Functions, Pub/Sub

Backend and Integration: FastAPI, NestJS, TypeScript, REST APIs, service integration

Infrastructure: Docker, Terraform, GitHub Actions, CI/CD, Git/GitHub

Edge Data and IoT: MQTT/Mosquitto, Home Assistant

EXTRACURRICULAR ACTIVITIES

Speaker

ECHO: Monitoring and Automation Platform for Remote Stations

Ingeniería para Chile Conference, PUC
May 2026

LANGUAGES

Spanish: Native

English: B2 / Upper-intermediate proficiency

ADDITIONAL INFORMATION

Portfolio

pedrozavalat.github.io

LinkedIn

linkedin.com/in/pedro-pablo-zavala

GitHub

github.com/pedrozavalat

REFERENCES

Rodrigo Arnaldo Carrasco Schmidt

Pontificia Universidad Católica de Chile
rcarrass@uc.cl